

LLMs: the Fundamentals

- Transformer architecture
- **GPT** in ChatGPT

Learning Goals

Describe how an attention layer works

Transformer

Generative

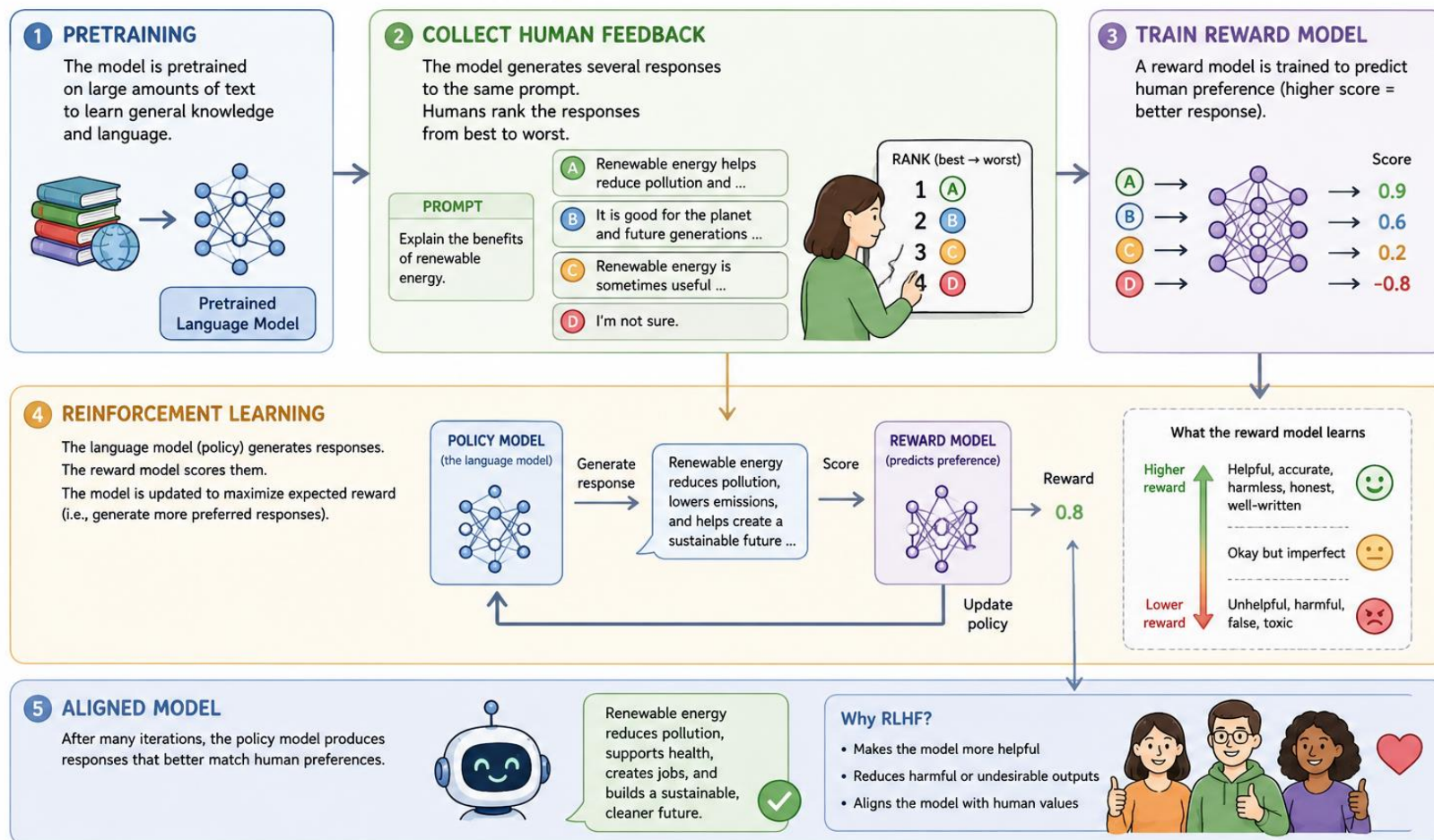
Pre-trained



The Alignment Problem: mitigated with Reinforcement learning from human feedback (RLHF)

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)

Aligning an AI model with human preferences



Note: In practice, RLHF also includes safety data, filtering, and iterative data collection.

Ai + Human >> Ai

[nature](#) > [news feature](#) > [article](#)

NEWS FEATURE | 02 July 2025

AI ‘scientists’ joined these research teams: here’s what happened

Emerging ‘co-scientist’ systems use chatbots to mimic the deliberations of a research group. *Nature* asked researchers to test them out.

By [Nicola Jones](#)

nature

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Article | [Open access](#) | Published: 25 March 2026

Towards end-to-end automation of AI research

[Chris Lu](#), [Cong Lu](#), [Robert Tjarko Lange](#), [Yutaro Yamada](#) , [Shengran Hu](#), [Jakob Foerster](#), [David Ha](#)  & [Jeff Clune](#) 

[Nature](#) **651**, 914–919 (2026) | [Cite this article](#)

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The More You Automate, the Less You See: Hidden Pitfalls of AI Scientist Systems

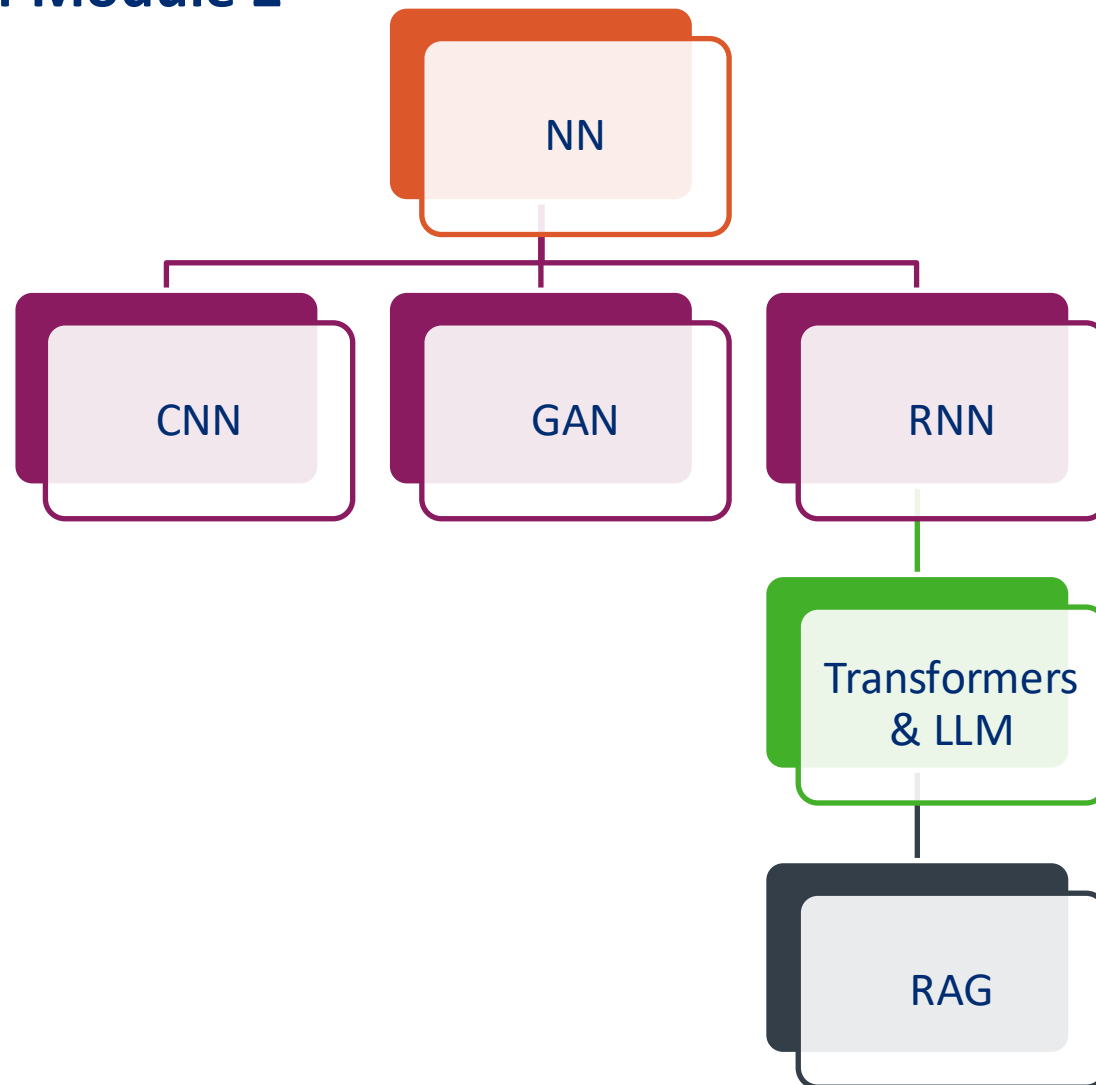
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Learning Goals

- Explain Transformer architecture
- Define “Generative – Pre-trained – Transformer” (GPT) terminology.
- Demonstrate tokenization & embeddings through live games (Semantis).
- Identify strengths & limits of large models in different modalities (text vs. code vs. images).

ChatGPT - Review of Module 2



Summary:

1. The goal of machine learning is to **learn patterns in data** and apply patterns **to predict** based on input

2. Neural networks are a type of machine learning that allow arbitrarily complex relationships to scale

- A bunch of regressions stacked together
- Billions of parameters

Today: Transformers

- **Self-Attention:** Global Pairwise Attention compared to CNNs (small local window)
 - Every token talks to every other token and considers every token pair (i, j)
- Captures long-range information: **No locality bias;** Allows context
 - CNNs only look in neighborhood; RNNs only 'see' past context step by step

Transformers

When do you need context? **JOKES**

An English professor, a physicist, and a statistician go out hunting. They spot a deer. The English professor writes a beautiful essay about the deer's elegance. The physicist carefully calculates the exact angle and muzzle velocity needed. The statistician says, "Let's assume the deer is in the crosshairs—then we'll see if we can hit it."

Transformers

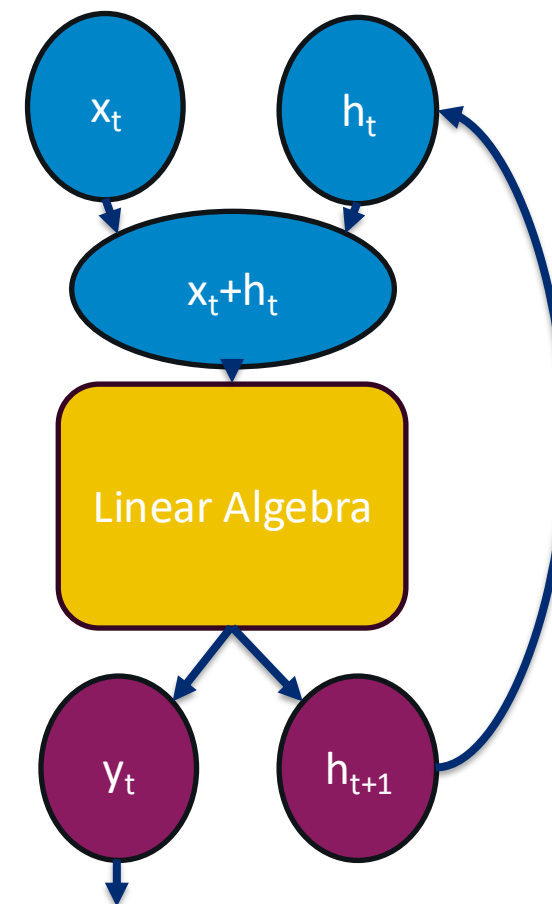
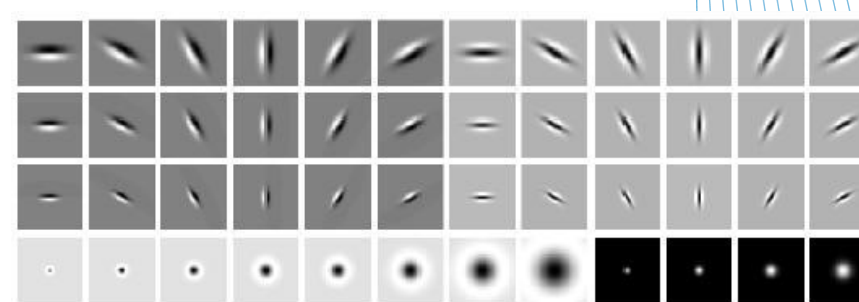
Two hunters are out in the woods when one collapses... He doesn't seem to be breathing, and his eyes are glazed. The other guy whips out his phone and calls the emergency services. He gasps, "I think my friend is dead! What can I do?" The operator says "Calm down. I can help. First, let's make sure he's dead." There is a silence, then a gun shot is heard. Back on the phone, the guy says "OK, now what?"

The Transformer & Self-Attention

Working with Sequences

Context is king

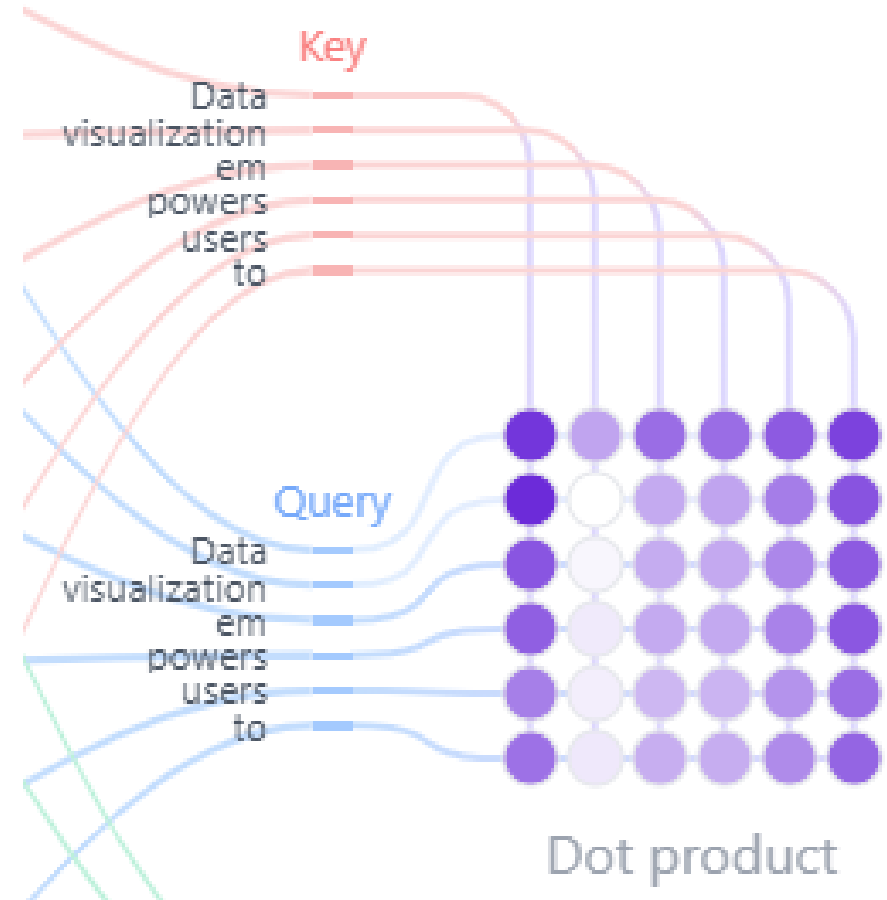
- What do convolutions and recurrent layers have in common?
- Capture context more efficiently than fully connected layers.
- Convolutions summarize data by finding patterns.
- Recurrence summarizes sequences into a single data point via cumulative memory.
- Both focus on local context



How do we broaden our context window?

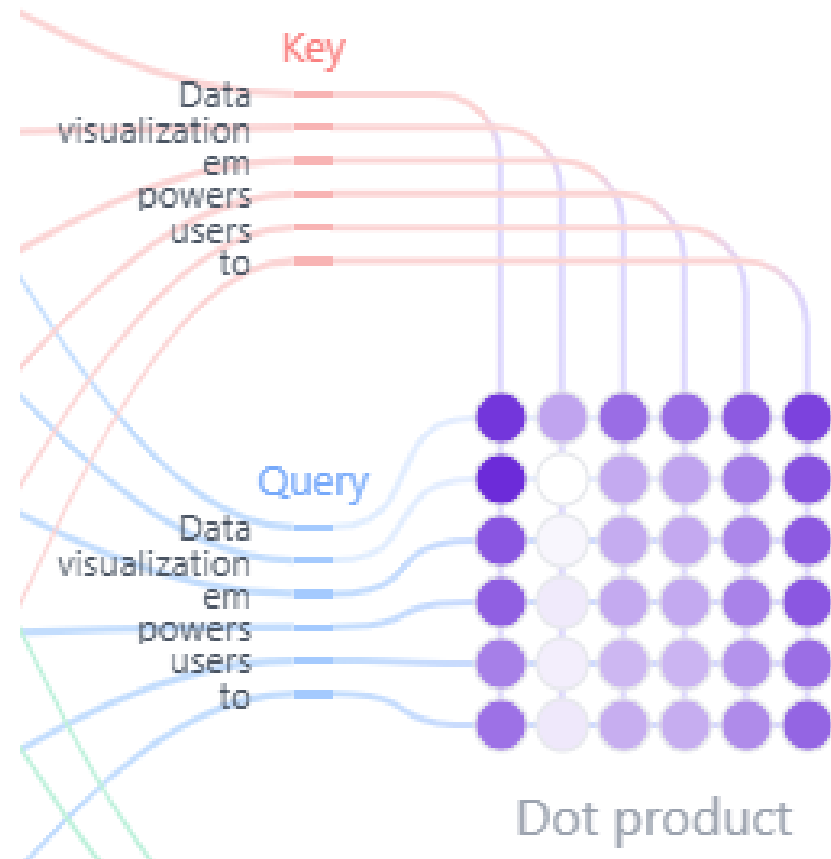
Self attention captures the relationships between input tokens and has 3 steps:

1. Measure the similarity between tokens
2. Scale the similarity using learned weights
3. Apply an activation function & regularize



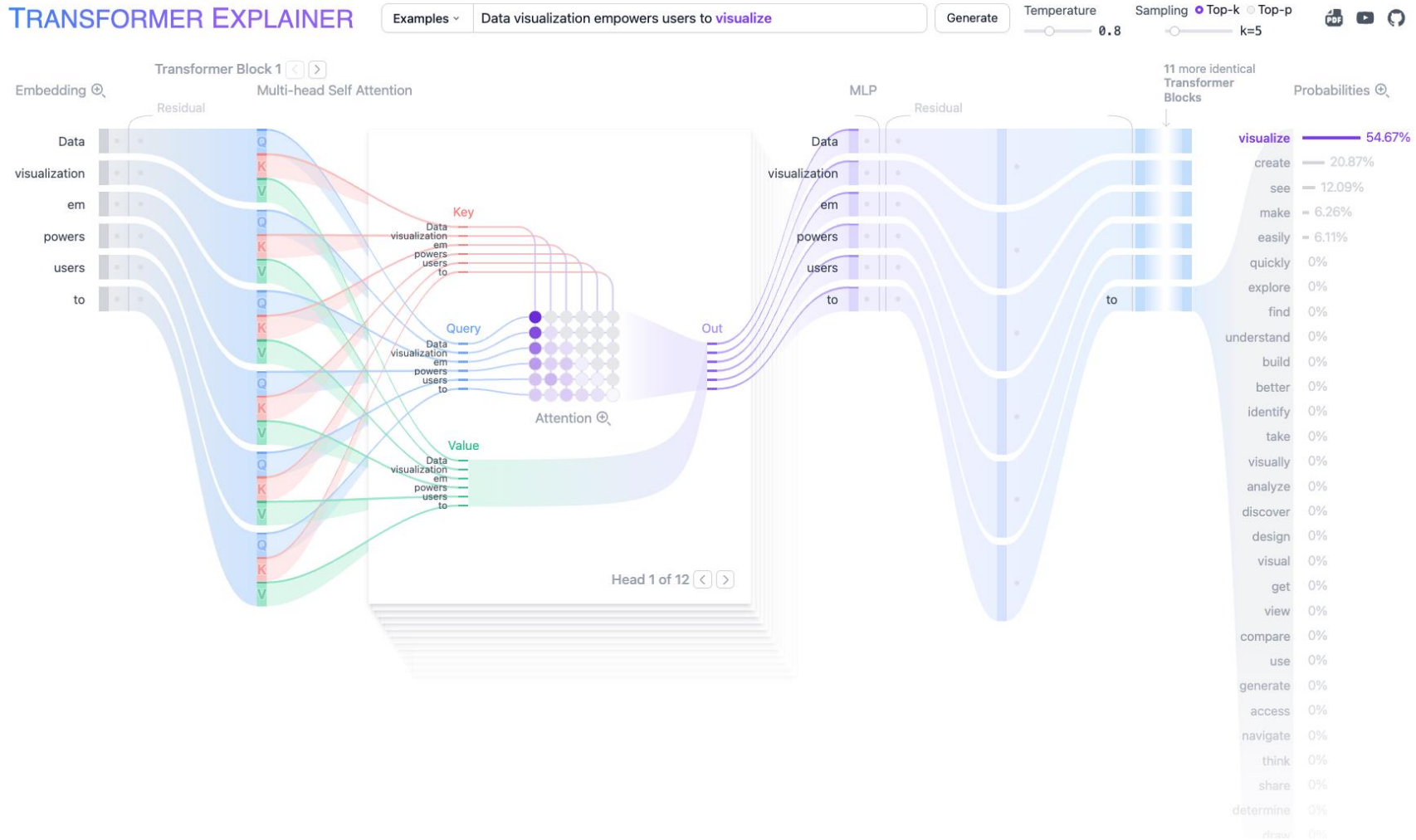
What are the advantages of self-attention?

- Can connect separated words if they are contextually related
- Multiple “attention heads” can capture different relationships
- Each attention head can be calculated in parallel



How do transformer work in practice?

TRANSFORMER EXPLAINER



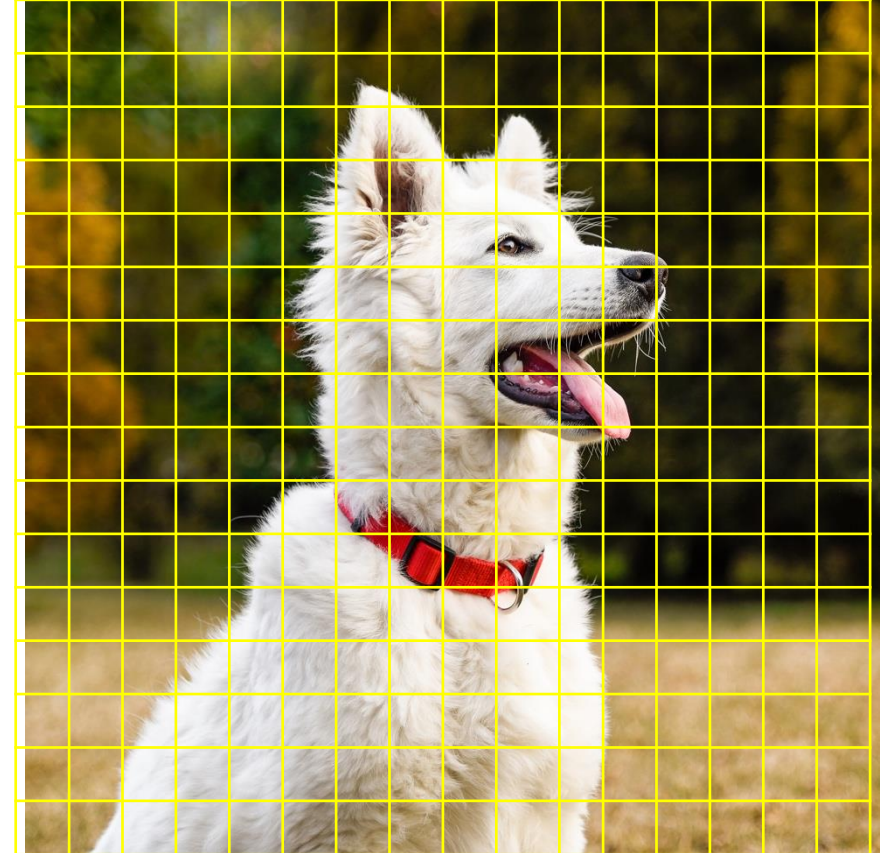
<https://poloclub.github.io/transformer-explainer/>



Brief aside: Tokens aren't just for words

Vision transformers use the same self attention mechanism to improve image analysis

- Much like tokenizing a sentence an image is broken down into patches
- Convolutions are used to turn each patch into a feature vector
- The feature vectors are used as tokens by the attention layer



Limitations of self-attention

1. Computationally expensive

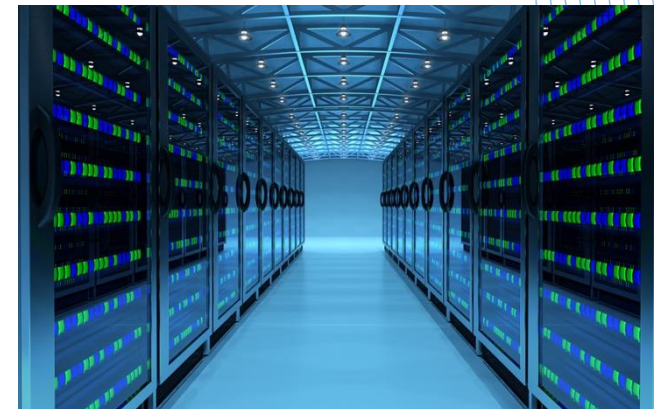
- While they are more parallelizable than RNNs transformers tend to take more total compute resources

2. Memory footprint

- Larger models can have trouble fitting in working memory
- Running out of working memory significantly slows down computation!

3. Required training data

- Transformers tend to need much larger training sets to learn the complex interactions they are designed to capture

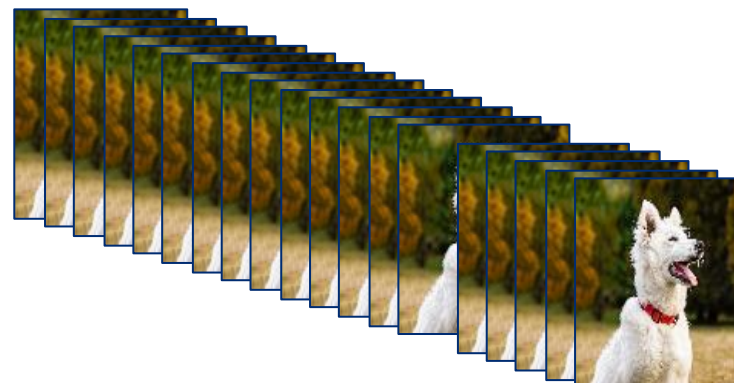


Computer Data

Working
Memory



Storage



What is a context window: What can LLMs ‘remember’?

- A context window is the amount of information that an LLM can actively consider at one time when generating a response
- Think of it as **Working Memory**
 - More context = better reasoning, summation, conversations
- It is not permanent knowledge
 - Information may be dropped

Human	LLM
Long-term Memory	Pre-training data
Working Memory	Context Window
Looking something up	Retrieval (RAG)

Transformers – Summary

- **Strengths:**

- Wide context
- Highly parallelizable
- Broadly applicable

- **Limitations:**

- Computationally expensive
- Memory footprint
- Requires training data

To discuss *after* activity

We will be replaying the sentence game from yesterday but this time the words are ordered based on “importance” to the sentence

1. How did this compare to analyzing the sentence sequentially?
2. Does this way of ordering things introduce any new concerns?

Sentence Analysis Activity

1. Break into small groups
2. Each group gets a stack of index cards
 - The top card is a random word from the target sentence
 - The rest of the cards are the sentence written out in order of “importance”
3. Try and guess the meaning of the sentence from the top card
4. Flip each of the remaining cards over one by one and try to guess the meaning of the sentence in as few cards as you can

To discuss *after* activity

We will be replaying the sentence game from yesterday but this time the words are ordered based on “importance” to the sentence

1. How did this compare to analyzing the sentence sequentially?
2. Does this way of ordering things introduce any new concerns?

To discuss *after* activity

- How much context do you need to understand the meaning of a word in a sentence?
- Do you think there's a difference between how humans and machines understand language? If so, what is it?
- What might be the trade-offs between looking only at nearby information versus everything at once?
- In real-world AI systems (like translation or chatbots), why might it matter whether the model is using local or global attention?

Match the Mindset

Let's review!



- 
- A. Supervised learning** (e.g., labelled regression/classification)
 - B. Regularization techniques** (dropout, weight decay)
 - C. Over-fitting** (plus guardrails like validation sets)
 - D. Unsupervised learning** (e.g., K-means clustering)
 - E. Linear/logistic regression**
 - F. Self-attention / Transformer**
 - G. Generative Adversarial Network (GAN)**
 - H. Convolutional Neural Network (CNN)**
 - I. Recurrent Neural Network (RNN)**
 - J. “k-fold cross-validation”**
 - K. Evaluation metrics** (accuracy, precision-recall, etc.)
 - L. Activation function** (ReLU, sigmoid, etc.)

BREAK

Come back at: XX:XX



Guest Speaker: Research IT

Ai at JAX



BREAK

Be back by XX:XX



ChatGPT



LLMs

ChatGPT

- Deep learning model
- **Generative:** predicts the next word
- **Pre-trained:** pre-trained for expectations based on massive amounts of text
- **Transformer:** type of neural network architecture (introduced in 2017, works by focusing attention onto relevant parts of the sentence)
 - We have learned about transformers today

How does a **L**arge **L**anguage **M**odel work?

Generative: predicts the next word

Q: Can we predict the missing words in the following sentence?

“Der schnelle braune fuchs springt _____”

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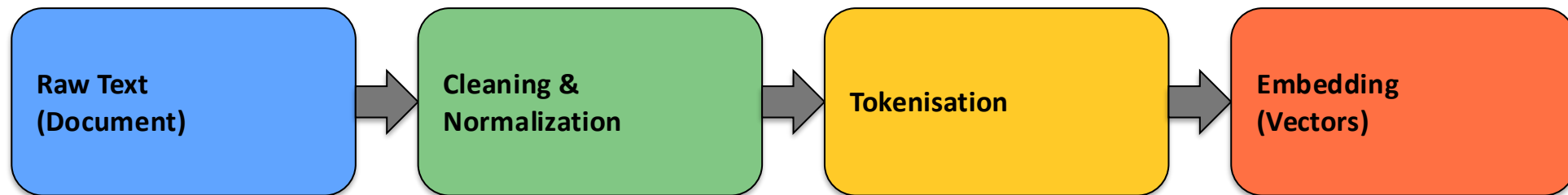
“The quick brown fox jumps _____”

How does a **Large Language Model** work?

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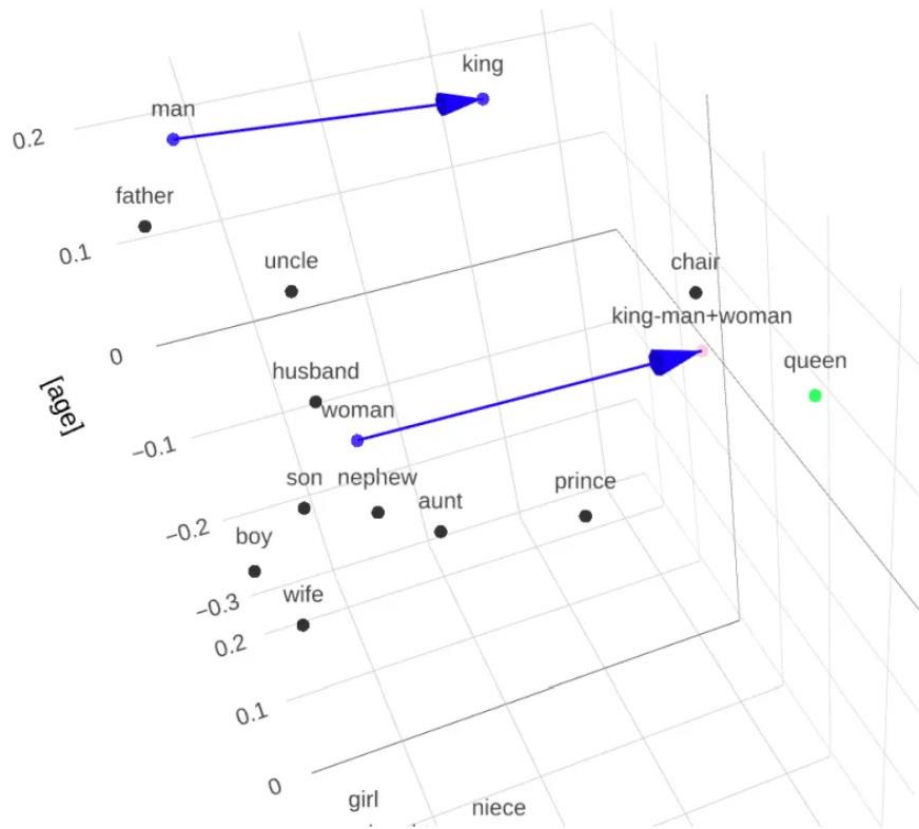
We are already familiar with the following:

Watch:

2-minute video on how Autocomplete works:
https://www.youtube.com/watch?v=us9tUY_yN7Y

- **Autocomplete:**
 - statistical n-gram (quantification of vector space of words)
 - 1–3-word context
 - Prediction target is the next word only

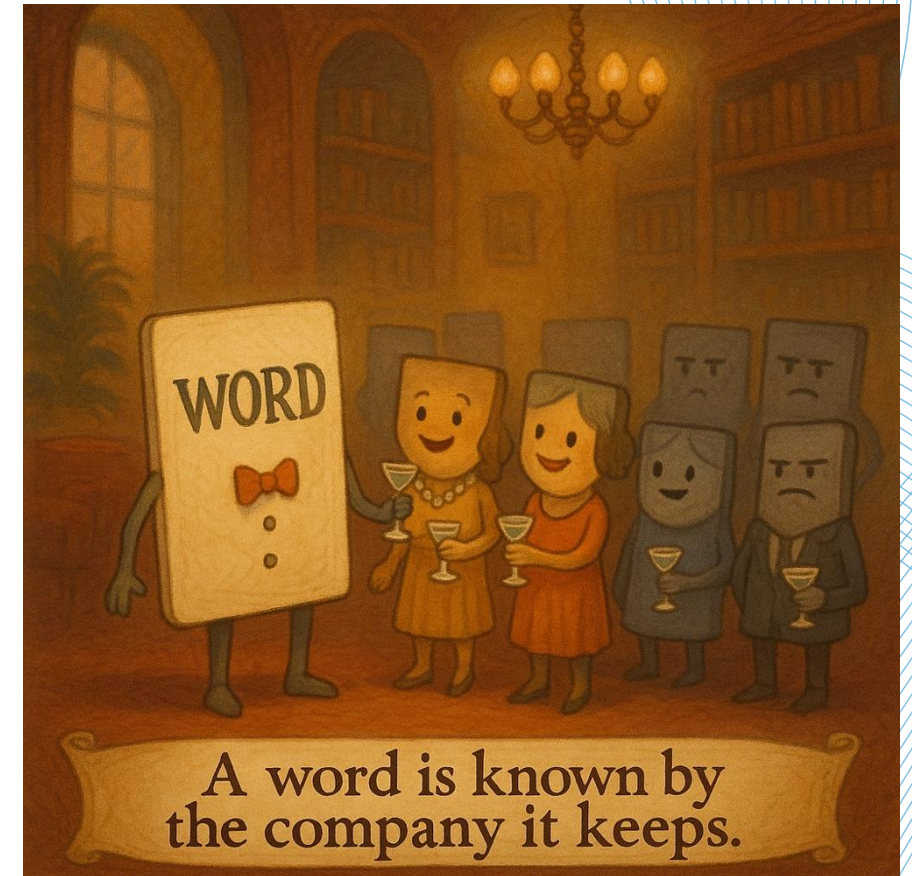
Vector and Semantic Space



“king – man + woman” = “queen”

Play:

<https://research.google.com/semantris/>



Vector and Semantic Space



How does a **Large Language Model** work?

Generative: mechanism is tokenization

Q: Can we predict the missing words in the following sentence?

“The quick brown fox jumps _____”

Play:

<https://research.google.com/semantris/>

More Challenging:

<https://semantle.com/>



How does a Large Language Model work?

- Additional practice: <https://www.hooplaimpro.com/mind-meld>
- **N-grams:**
- Precursor to LMMs
 - **Trigram:** $P(w_i | w_{i-2}, w_{i-1})$
 - Counts how often the three-word sequence, (w_{i-2}, w_{i-1}, w_i) , occurs
 - Captures dependencies within the three-word window
 - Problem: rare words
- Demo: <https://www.cs.cmu.edu/~dst/MarkovChainDemo/>

“The best thing about AI is its ability to...”

The best thing about AI is its ability to

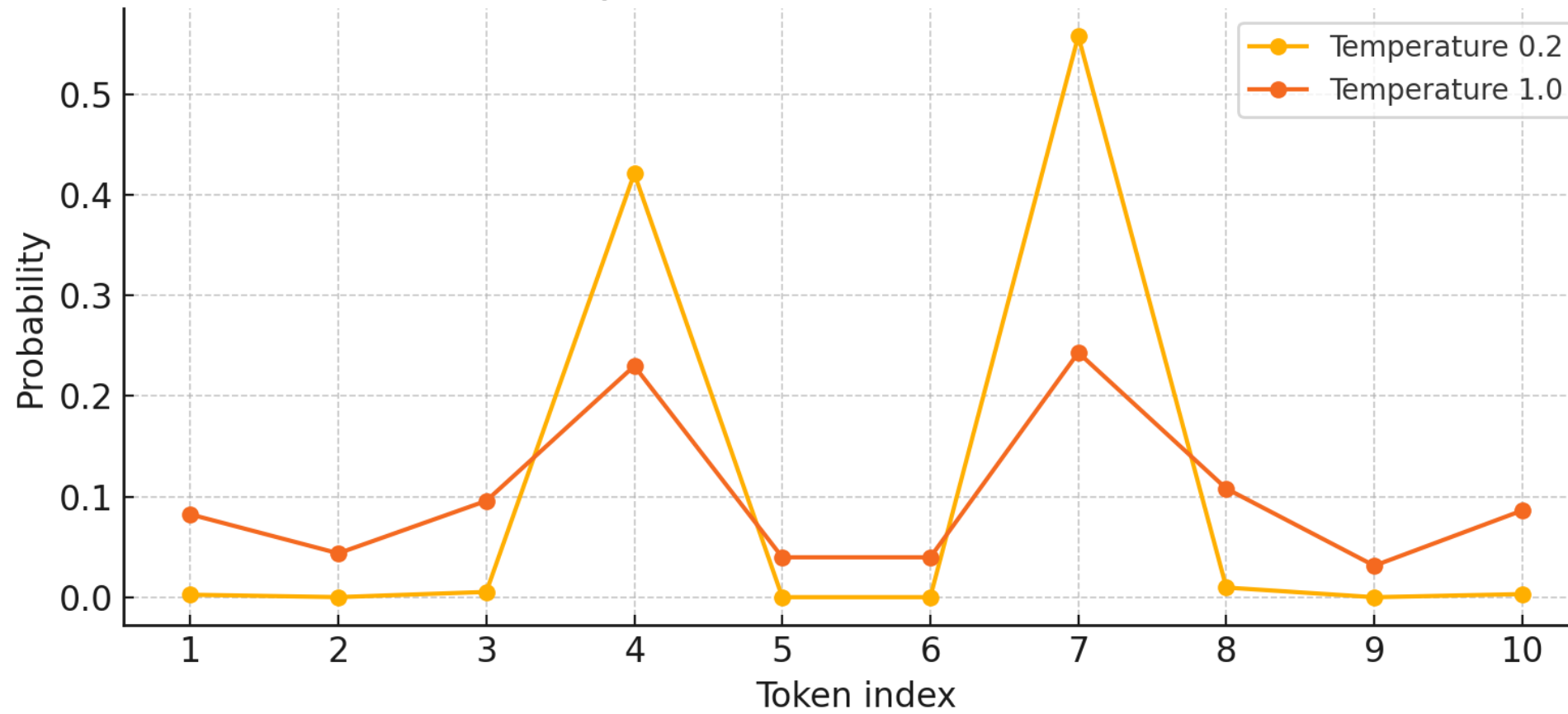
learn	4.5%
predict	3.5%
make	3.2%
understand	3.1%
do	2.9%

Stochastic aspect:

- randomly doesn't pick the highest probability word ('temperature'=0.8)
- Same prompt can result in different results
- This is a ML classification task

“The best thing about AI is its ability to...”

Effect of Temperature on Next-Token Distribution



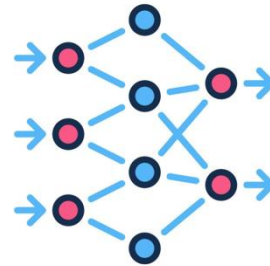
* Many common models don't allow you to adjust this manually, just implicitly in your phrasing



“The best thing about AI is its ability to...”

The best thing about AI is its ability to

learn	4.5%
predict	3.5%
make	3.2%
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do	2.9%



{ The best thing about AI is its ability to,
The best thing about AI is its ability to create,
The best thing about AI is its ability to create worlds,
The best thing about AI is its ability to create worlds that,
The best thing about AI is its ability to create worlds that are,
The best thing about AI is its ability to create worlds that are both,
The best thing about AI is its ability to create worlds that are both exciting,
The best thing about AI is its ability to create worlds that are both exciting, }

Because of the random choices, the same input prompt can result in different outputs:

The best thing about AI is its ability to learn. I've always liked the

The best thing about AI is its ability to really come into your world and just

The best thing about AI is its ability to examine human behavior and the way it

The best thing about AI is its ability to do a great job of teaching us

The best thing about AI is its ability to create real tasks, but you can

ChatGPT

Let's see what ChatGPT will provide us as answers to the following questions:

1. The technical definition of a large language model (LLM) is:
2. Large language models (LLMs) can be used to create a variety of different outputs, such as:

Let's compare these answers to the ones provided by Google Search autocomplete

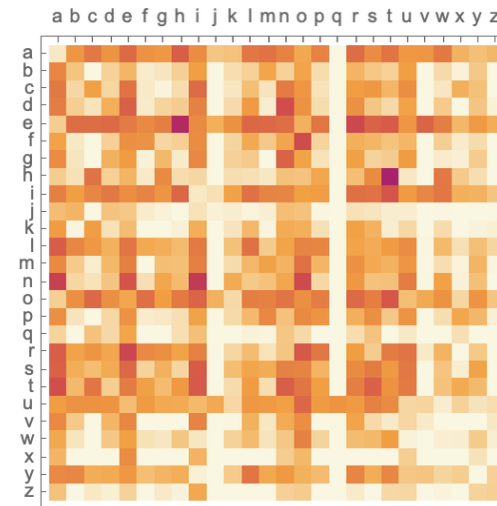
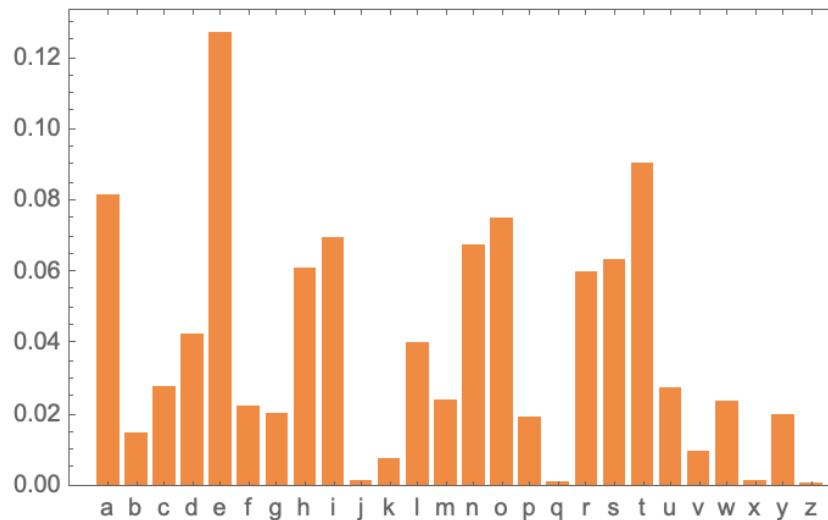
LLMs

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Pre-training in ChatGPT

1. Cryptography has single letter and duo letter probabilities (2-grams) for English.



2. Training on a large corpus of books, you can get probabilities for words, but you need a model that has parameters: billions of knobs to turn

Pre-training in ChatGPT

- LLM are trained on huge amounts of data
- Allows them to predict next token
- Requires huge resources



VS



Pre-training in ChatGPT

Different types of Training

1. Pre-training (self-supervised): Knowledge

- Fed massive amount of information, but this only allows to babble (continuation), not **alignment**
- Not real 'learning' just regurgitating, like a fancy Google search autocomplete on steroids
- **Best Example: text summation**
- "Summarize this article in one sentence"

2. Instruction fine-tuning (supervised): Format

- models learn to respond, begin to **align**
- **High quality examples of instruction and response pairs**
- **Best Example: Answering questions**
- "In two sentences, compare Newtonian gravity and Einstein's general relativity"

3. Reinforcement learning (from human feedback): Judgement

- helps **alignment**
- **Critical for improving performance**
- **Human rate answers which is then used in subsequent iterations**

LLMs

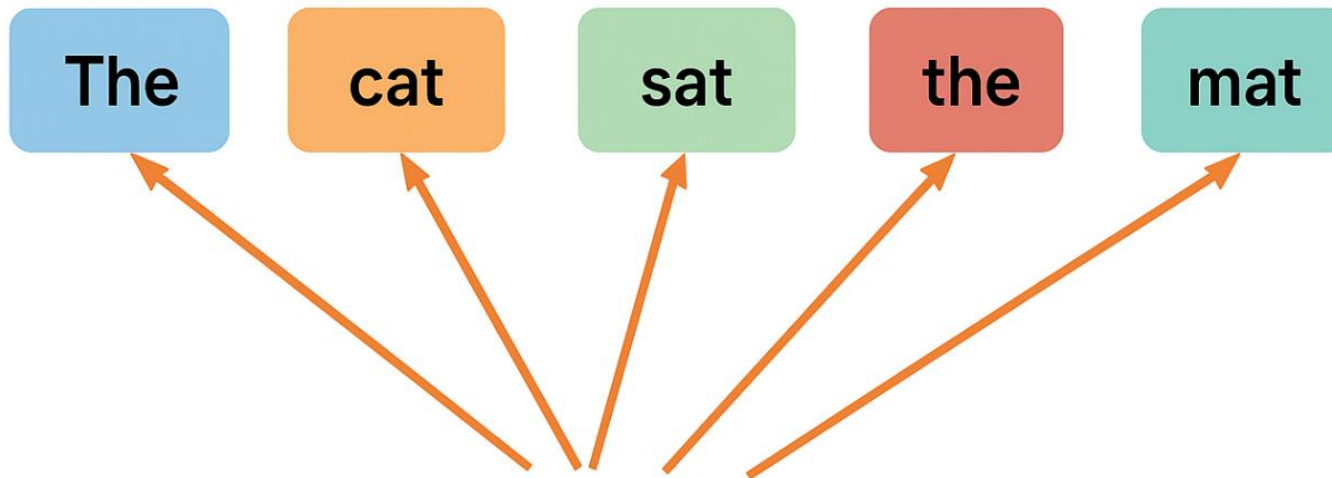
ChatGPT

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Transformers:

Self-attention: Global Pairwise attention

- every token talks to every other token and considers every token pair (i, j)
- Captures long-range information: **No locality bias**; allows context



Self-attention: “cat” attends to every other word

How do we evaluate the output of LLM?

Human-in-the-loop

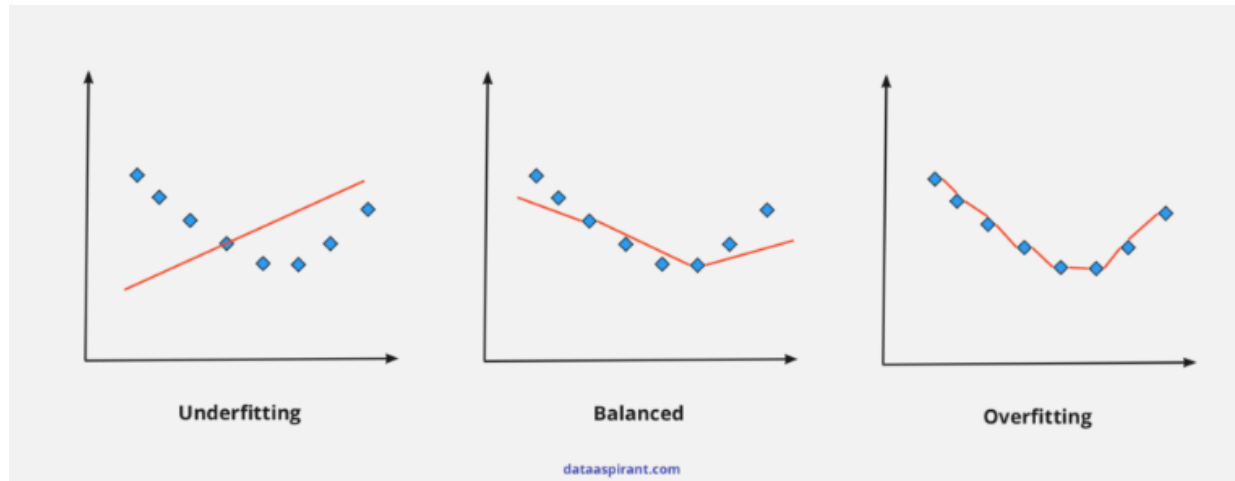


Human-in-the-loop

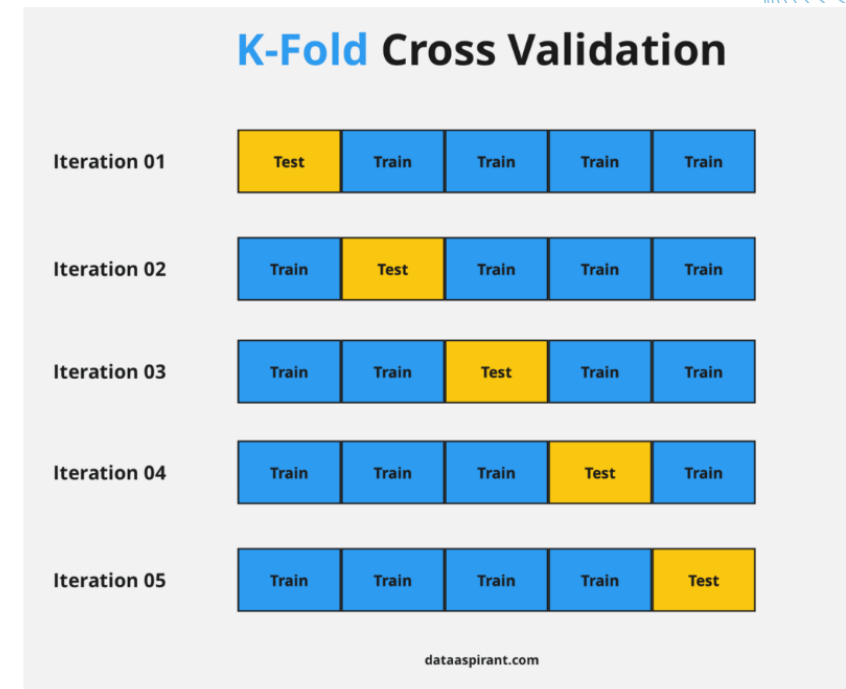
- **Training** versus **Testing**

Strategy:

- split the data and use some of it for **training** the NN to **make predictions** and **reserve** some of it **to test** the NN



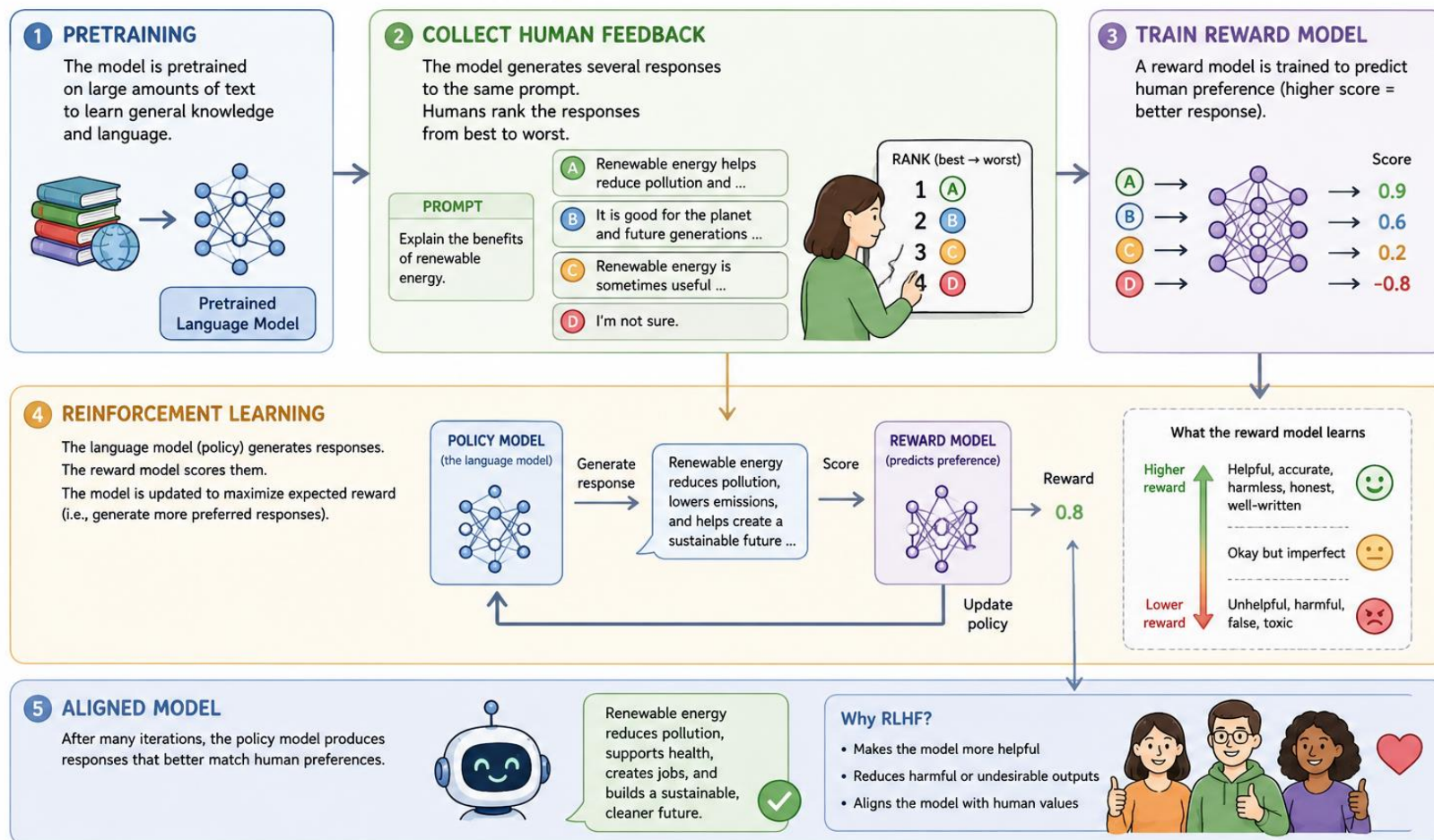
- K-fold cross-validation



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Aligning an AI model with human preferences



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Human-in-the-loop

Stroke: Vascular and Interventional Neurology



EDITORIAL ▶ Stroke Vasc Interv Neurol. 2026 Mar 3;6(2):e002321. doi: [10.1161/SVIN.126.002321](https://doi.org/10.1161/SVIN.126.002321)

Jevons Paradox: Rise of the Machines May Create More Work for Physicians

[Jens Fiehler](#) ^{1,8}

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PMCID: PMC12959442 PMID: [41815308](#)

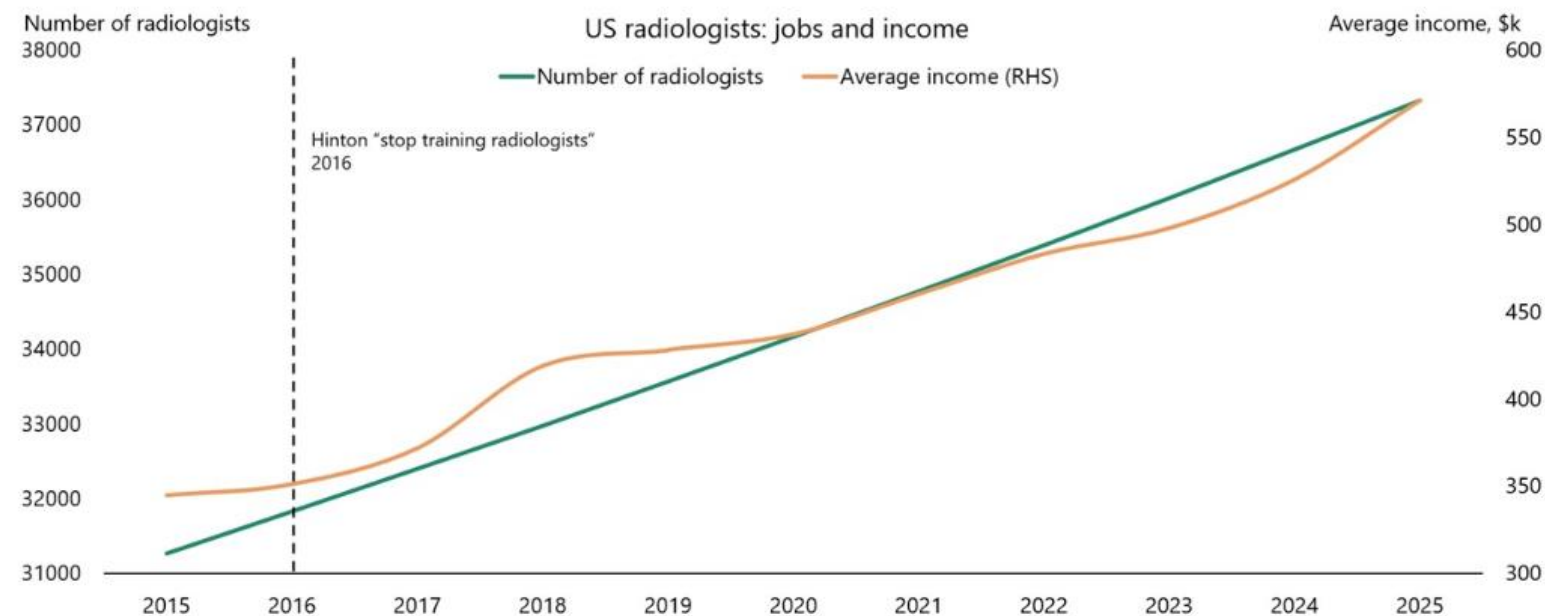
*It is a confusion of ideas to suppose that the economical use of fuel is equivalent to diminished consumption. The very contrary is the truth.*¹

Stanley Jevons (1865)

The surprising thing is the prediction that radiologists would be the first jobs to go was exactly the opposite. The trend shows that more radiologists are being hired now as a result of AI.

Jensen Huang, CEO, Nvidia, November 19, 2025, U.S.-Saudi Investment Forum²

The radiologist paradox



Sources: CMS Provider Data Catalog (National Downloadable Files), endpoints 30,723 (2014) and 36,024 (2023) per Rosenkrantz et al., AJR 2024; Medscape Physician Compensation Reports 2016–2026, Apollo Chief Economist